Matias Barandiaran

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Education

Lancaster University Leipzig, Germany
Bachelor of Science in Computer Science October 2022 – July 2025

- Ranked **1st** out of **212** students.
- First Class Honours (**UK**: 91.2% ≈ **GPA**: 4.0/4.0).
- University ranked **#132** worldwide for Computer Science (QS by subject) in 2025.

San Ignacio de Recalde School Lima, Peru
International Baccalaureate March 2020 – December 2022

- Cohort's top 5% (**GPA**: 3.9/4.0).

Academic Gaps

August 2025 – Present: Prepared Master's application, presented at International Conference on Artificial Life 2025, enrolled in German language lessons.

January 2022 – September 2022: Transition period between end of Peruvian school year and start of UK undergraduate year. Completed visa processing, relocation, and academic preparation.

Work Experience

Technical University of Munich Munich, Germany (Hybrid)
Research Assistant (Hiwi) July 2024 – August 2025

- Hosted by the Gagneur Lab and Helmholtz Munich.
- Worked on rare variant association testing using deep learning and data-driven burden scores (DeepRVAT), utilizing **Python**, **R**, **PyTorch**, and **Snakemake**.

Martin Luther University of Halle-Wittenberg Remote
Research Intern July 2024 – September 2024

- Hosted by the Faculty of Medicine.
- Worked on graph-level anomaly detection via latent space searching and generative adversarial networks (gladGAN) on molecular datasets, utilizing **Python** and **Pytorch**, and **scikit-learn**.

Lancaster University Leipzig Leipzig, Germany
Teaching Assistant October 2023 – June 2024

- Delivered instructional sessions of **MIPS** assembly programming for the Digital Systems module.

Charles University Prague, Czechia
Research Intern August 2023 – September 2023

- Hosted by the Faculty of Mathematics and Physics.
- Contributed to: *Unsatisfiability Proofs for Horn Solving* (2025). *ETAPS best paper award*
- Designed production of Alethe verifiable proofs for the GOLEM horn clause solver in **C++**.

Freelance Lima, Peru
Math Tutor August 2020 – September 2021

- Guided senior high school students to achieve outstanding exam performance and substantial growth in their mathematical abilities (IB Mathematics SL and HL).

Publications

Barandiaran, M. & Stovold, J. (2025). Growing Reservoirs with Developmental Graph Cellular Automata. In Proceedings of International Conference on Artificial Life 2025 (pp. 26–35). MIT Press.

Otoni, R., Blicha, M., Barandiaran, M., Eugster, P., Kofrŏn, J., & Sharygina, N. (2025, May). Unsatisfiability Proofs for Horn Solving. In Proceedings of International Conference on Tools and Algorithms for the Construction and Analysis of Systems (pp. 67-87). Springer Nature.

Awards / Honours

Chancellor's Medal – Lancaster University (2025): The university's most prestigious award, presented annually to graduating students who demonstrate exceptional merit in their field.

Valedictorian – Lancaster University (2025): Ranked **1st** out of **212** students achieving the cohort's highest average grade across all three years of the BSc Computer Science program.

Outstanding Dissertation – Lancaster University (2025): Obtained the highest achievable grade (**100%**) for the BSc Computer Science dissertation: *Growing Reservoirs with Developmental Graph Cellular Automata*. **Strong Accept** for presentation at peer-reviewed International Conference on Artificial Life 2025 in Kyoto, Japan.

EASST Best Paper Award – Charles University (2025): Best ETAPS (International Joint Conferences On Theory and Practice of Software) 2025 paper related to the systematic and rigorous engineering of software and systems.

Scholarship – Lancaster University (2022): Awarded a €4,000 tuition scholarship in recognition of academic merit.

Leadership Activities

Lancaster University Leipzig, Germany
Founder, **Computer Science Society** 2023 – 2025

- Dedicated to fostering a culture of innovation, diversity and problem-solving.
- Secured university funding for the purchase of robotics supplies (e.g., 3D printer, Arduinos, etc.).

Lancaster University Bailrigg, England
Co-Organizer, **Global Advancement Fund** 2024

- Co-Initiated and championed the approval process of a ~€20k fund to facilitate international collaboration between Lancaster campuses worldwide.

Lancaster University Leipzig, Germany
Lead Organizer, **Qiskit Fall Fest** 2023

- Quantum Computing (QC) event powered by IBM.
- Gave an introductory talk on QC and held workshops with Qiskit material. Assembled a series of seminars with special guests like Enrique Solano (Co-CEO of Kipu Quantum).

Projects

Industrial Waste Sorting Arm – Lancaster University (2024): Integrated an industrial robotic arm with object detection and image classification software for the purpose of automated waste sorting, utilizing **Python**, **PyTorch**, **C++**, **C**, and **OpenCV**. Funded by Lancaster University's Global Advancement Fund.

Global Game Jam – Lancaster University (2023, 2024): Participated in 48-hour game development events using **Java** and **libGDX**. Collaborated in a team using **agile development** and **continuous integration** practices.

Certifications

AWS Certified AI Practitioner – Amazon Web Services (2025): Demonstrated proficiency in applying foundational AI/ML concepts, model deployment workflows, and cloud-native AI services within AWS.

Prompt Engineering for Developers – DeepLearning.AI & OpenAI (2025): Covered techniques for designing effective prompts, controlling model behaviour, and building production-grade LLM applications through the OpenAI API.

Advanced Applications of Neural Networks and Deep Learning – University of Oxford (2023): Explored Generative Adversarial Networks, Variational Autoencoders, Deep Reinforcement Learning, Diffusion Models, and Graph Neural Networks.

Artificial Intelligence and Machine Learning: Theory and Practice – University of Oxford (2023): Studied the underlying mathematical foundations of linear and convolutional neural networks.

Additional Skills and Interests

Languages: Spanish (native), English (C2) ***TOEFL**: 114/120*, German (A2.1).
Tools: Python, Java, R, PyTorch, Git, C, C++, SQL, Linux, Snakemake, QEMU, Amazon Bedrock.
Concepts: Rare Variant Association Testing, Data Pipelines, Computer Vision, Generative Modelling, Supervised Learning, Reinforcement Learning, Reservoir Computing, Cloud Computing.
Hobbies: Game Development, Fishwatching, Badminton.